

INAL EXAM

Fall 2019

- This is a closed-book exam.
- Students whose native language is not English may use a dictionary if they wish.
- Please check that you have pages in your copy of the exam.
- There are 180 points in the exam. You also have 180 minutes.
- **Please put the first section, covering the shorter questions, in one book and the longer questions in a second book. Label all books.**
- Remember to put your name on all books.
- Do not be concerned if your answers are not whole numbers.
- You cannot use your phone as a calculator. Simple calculators will be provided by AASL office, and are sufficient for completing all calculations in the exam.
- If your answer is a number put a box around it. Example: P=\$436, Q=35,465.

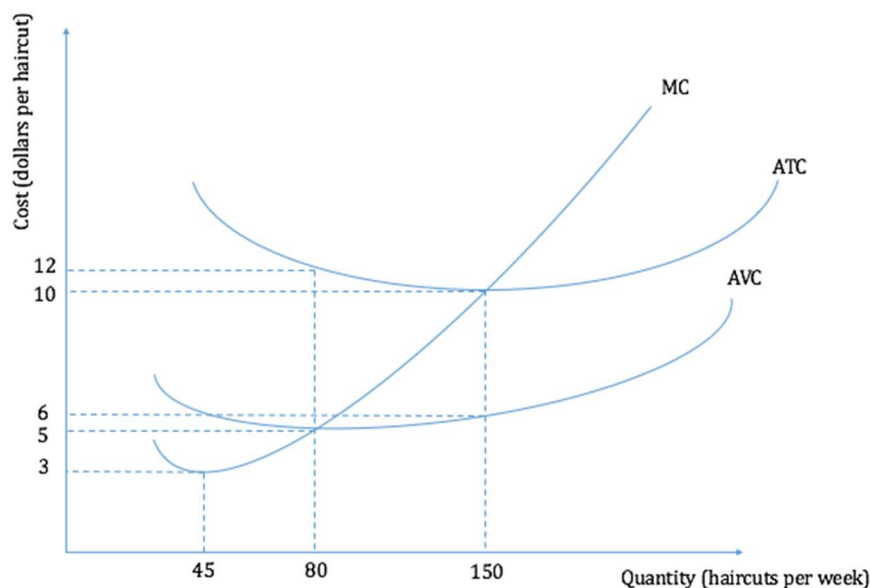
Section I: Short Questions [60 points]

Please be concise. None of the answers requires more than a brief paragraph.

1. The city of New Haven plans to impose a tax on rental housing. Under the proposal, landlords will be required to pay a 10 percent tax on all rent received. The Board of Alders (the local government officials who will vote on the proposal) decides that landlords are pretty greedy and since the tax is on them and not on tenants, the proposal sounds pretty good. Is their reasoning sound? Why or why not? (You can use a diagram to supplement your answer, if you want.)

No, they are wrong. Incidence is shared and depends on relative elasticities of supply and demand. Usual diagram showing tax incidence.

2. The business of basic haircuts is often described as perfect competition, since there are few barriers to entry, the product is undifferentiated, and there is an ample pool of labor qualified to give haircuts. The diagram below shows marginal cost, average variable cost, and average total cost for a business offering basic haircuts in New Haven. What do you expect the long-run average price to be for a basic haircut in New Haven, and **why**?



Answer: Average price will be \$10;
Entry and exit will keep the price at 10.

3. Imagine that you are a monopolist provider of specialized salon services in a small town in upstate New York. (Assume, that demand for salon services is linear, and you have a constant marginal cost of providing the services.) As a result of new legislation, the minimum wage in NY increases, which means that the marginal cost of your salon services increases. You tell your board members that because you are a monopolist, you should just pass on the full extent of the

increase in your costs to your customers by an equivalent price increase. Assuming that you continue to maximize profits, is your claim to your board members correct? Explain why or why not.

As a monopolist, you will optimally pick price and quantity, given the demand curve you face. In particular, you choose the quantity at which your marginal revenue equals marginal cost. If your marginal cost increases, you will re-optimize and choose a new price-quantity pair. With linear demand and constant MC, it will not be optimal to increase prices by the same amount as the cost increase. If you draw a picture this will become clear. Your optimal choice given the new MC will involve increasing the price by less than the full cost decrease.

4. A new spin studio is opening in East Rock, and they have decided they face a constant marginal cost per class attended. They have done some market analysis and have concluded that they can maximize profits by offering a “menu” of classes to New Haven spin enthusiasts. In particular, they found that the optimal menu was a package of 5 classes per month for \$75 (“light” package), and a package of 12 classes per month for \$150 (“fanatic” package). This menu would attract the two main segments of customers they identified, “light users” and “spin fanatics”. However, soon after the analysis was completed, Yale SOM announced it was growing the full-time MBA program, which meant more “spin fanatics” would be in the market than expected. These new “spin fanatics” have the same monthly demand curves as existing “spin fanatics”. Considering the size of the “fanatic” package (12) and its price (\$150), which might you expect to change in response to this, and in which direction (increase or decrease)?

Answer: Fanatic size does not change; fanatic price increases.

5. A coworker who has been to business school said that the problem with perfectly competitive markets like shipping or construction is that no firm in a perfectly competitive market can ever earn a profit. Is this always true? Do you agree or disagree and **why**?

Answer: Agree “in the long run”; however, in the short run, not true: very possible to earn profits in the short run. This is why firms enter when profits are high.

6. Uber is known to be constantly experimenting with pricing. They recently increase the price of a trip from New Haven to Bradley airport from \$70 to \$77. They found that the number of rides completed fell by 15%, while Lyft saw its number of rides increase by 3%. What can you (**precisely**) conclude about Uber’s own-price elasticity of demand and the cross-price elasticity of demand for Lyft with respect to Uber’s price on this route?

Answer: Uber faces elasticity of -1.5; Lyft cross price elasticity is 0.3 and is a substitute.

Section II: Long Questions [120 points]

Problem 1: Bonuses at Deutsche Bank (16 points)

Times are tough at Deutsche Bank in New York. The investment banking team is worried about bonuses,

especially as several deals are looking shaky. Consider a banker whose base salary is \$400K per year, and who is expecting a bonus of \$200K if the T-Mobile/Sprint merger is blocked, but a bonus of \$800K if it actually goes through. Current odds have the merger approval happening at 50%. This banker has utility over wealth of $u(x) = \sqrt{x}$ with x denominated in thousands, so that the utility of 100K is $\sqrt{100} = 10$. Suppose she also has outside savings of \$500K.

- a) (4 points) What is the **expected value** of her bonus, and of her wealth at the end of the year?

Answer: bonus: $0.5 \cdot 200k + 0.5 \cdot 800K = \$500k$,
wealth = $500K + 400K + 500K = 1.4M$

- b) (5 points) Her boss has offered her a deal: given all this uncertainty, she could trade away this uncertainty for a guaranteed bonus of \$400K. Would she take it? (expected utility should include all wealth at the end of the year: base salary + savings + bonus).

Answer: $EU(\text{sure thing}) = \sqrt{500 + 400 + 400} = \sqrt{1300}$
 $EU(\text{risk}) = .5\sqrt{900 + 200} + .5\sqrt{900 + 800}$

The certainty equivalent of the risky bet is given by c where $.5\sqrt{900 + 200} + .5\sqrt{900 + 800} = \sqrt{c}$
This solves to 1384K.

She is being offered 1300K. So she should take the risk, and not take the deal!

- c) (7 points) What is the lowest deal she would take to trade away the 200k-800k bonus risk?

Answer: We solved that the risk is worth a certainty equivalent of 1384K.
Her savings and base salary give her a sure amount of 900, so she would demand a guaranteed bonus of at least 484K to trade away the risk.

Problem 2: Amazon Prime memberships (16 points)

Amazon has two distinct market segments for its Amazon Prime membership. Students ("S"), who do a lot of their shopping on Amazon but who don't have much disposable income, and families ("F"), who have more disposable income. Demand for prime memberships for each subgroup (in thousands) is given by:

$$Q_S = 250 - 5P_S$$

$$Q_F = 630 - 3P_F$$

Amazon estimates it has a constant marginal cost of \$30 per prime membership.

- a) (6 points) What is the profit-maximizing price for a Prime subscription to each group?

Answer:
For students:
 $P = 50 - (1/5)Q$
 $MR = 50 - (2/5)Q$

MR=MC yields $Q=50$ and $P = \$40$.

For families

$$P = 210 - (1/3)Q$$

$$MR = 210 - (2/3)Q$$

MR=MC yields $Q=270$ and $P = \$120$.

- b) (6 points) Amazon is able to price discriminate in this way because it can verify the identity of students. This ability is new: previously, anyone with an .edu email address could claim to be a student and get the student price. So, anyone who worked for a university, or who had an alumni email address could claim to be a student. What price would they have charged for a membership prior to the introduction of the verification system, when they could not verify someone's student status and assumed that everyone, families and students alike, would sign up as a student? (i.e. because all families have at least one member with a .edu email address)

Total market demand

$$Q = 880 - 8P \text{ if } P < 50 \text{ and } Q = 630 - 3P \text{ if } P > 50$$

Now using aggregate demand, we have

$$P = 110 - (1/8)Q$$

$$MR = 110 - (1/4)Q$$

MR=MC yields $Q=320$ and $P=70$.

But notice that 70 is above the price at which any student would buy.

So the correct price to set in this case is still \$120. – the profit maximizing price for the family segment.

- c) (4 points) How much would Amazon have been willing to pay for the verification system?

Answer:

Under PD, profits are $(50 \cdot (40 - 30)) + (270 \cdot (120 - 30)) = 24.8M$.

Under no PD, Amazon would choose to serve only family segment and get profits of

$$(270 \cdot (120 - 30)) = 24.3M$$

So should be willing to pay 500K more to target student segment separately.

Problem 3: Bars in New Haven (16 points)

New Haven has a lively bar scene in the downtown core, with creatively named places like “BAR” providing pizza and drinks to the town's residents. Unfortunately, things have been getting a little extra rowdy lately, and the town would like to tax alcohol consumption in the downtown core to deter things from getting out of hand.

The supply and demand of alcoholic beverages in downtown New Haven are given by:

$$Q_S = 400P - 1000$$

$$Q_D = 3000 - 400P$$

- a) (4 points) What is the market equilibrium price and quantity prior to the imposition of any tax?

Answer:

$$\begin{aligned}P_S &= 0.0025Q + 2.5 \\P_D &= 7.5 - 0.0025Q \\0.0025Q + 2.5 &= 7.5 - 0.0025Q \\0.005Q &= 5 \\Q &= 1000 \\P &= 5\end{aligned}$$

- b) (4 points) Suppose the tax is \$1 per beverage. What is the new equilibrium price and quantity?

Answer:

$$\begin{aligned}0.0025Q + 3.5 &= 7.5 - 0.0025Q \\0.005Q &= 4 \\Q &= 800 \\P &= 5.50\end{aligned}$$

- c) (5 points) The town alders have some concerns about this proposed tax. They want to be certain that it i) reduces consumption by at least 10% to solve the rowdiness problem, but also that ii) consumer surplus falls by less than 30% to ensure they are re-elected. Would this tax meet both criteria i and ii?

Answer: yes, it reduces consumption by over 10%. CS goes from $(1/2)(1000)(7.5-5)=1250$ to $(1/2)(800)(7.5-5.5)=800$
NO (second requirement)

- d) (3 points) What percent of the tax revenue from this proposed tax is “paid” from consumer surplus?

Answer: 50%.

Problem 4: Mac Pro Assembly (16 points)

Apple assembles its made-in-the-USA “Mac Pro” product at a facility in Texas using a mix of labor (“l”) and machines (“m”). They have the following production function for each assembly line, where output is in 1000s of units per year:

$$f(m, l) = 2 \cdot m + 8 \cdot l + 0.2 \cdot m \cdot l$$

They pay workers a salary and benefits package that costs \$100K per year, while machines cost \$50K per year in maintenance and licensing fees from their manufacturer.

- a) (6 points) Apple is currently employs 15 people and 32 machines on its first assembly line. What is the output of the assembly line per year, and how much does Apple spend to produce that output?

Answer: $15 \cdot 100 + 32 \cdot 50 = 3.1\text{M}$ in cost,
to produce $2(32) + 8(15) + 0.2(15)(32) = 280\text{K}$ units

- b) (4 points) Apple wants to know if this is the cost-minimizing production plan. Compute the marginal productivity of investing (MPI) of each input (where MPI is the marginal product of the input scaled by the input price, in thousands). Could Apple do better?

Answer:

$$MPI_M = \frac{2 + 0.2l}{50} = 0.1$$

$$MPI_L = \frac{8 + 0.2m}{100} = 0.144$$

Should use more labor

- c) (6 points) Based on the production function and current input prices, can you come up with a general rule of how many workers Apple should employ, as a function of the number of machines it uses? In other words, I am asking for the optimal l^* as a function of m^* .

Answer:

$$\frac{2 + 0.2l}{50} = \frac{8 + 0.2m}{100}$$

$$200 + 20l = 400 + 10m$$

$$20l = 200 + 10m$$

$$l = 10 + 0.5m$$

Problem 5: Airline pricing (16 points)

A recent trend in airlines is the advent of “Basic Economy”, a fare class that comes without any of the luxury features of standard economy, like being able to choose a seat or use the overhead bins. These fares can easily be \$30-100 less than standard economy. As a pricing expert for an airline, you were concerned about frequent flyers defecting to purchase these lower basic economy fares, and so you also recommended that basic economy fares contribute nothing to “status” with the airline, don’t earn “miles” with the loyalty program, and that status perks such as upgrades do not apply. These fares have been introduced on many, but not all routes the airline operates.

Consider the potential market for the route BDL-ORD, Hartford to Chicago. **There are 5 casual flyers for each frequent flyer.** You have a constant marginal cost per customer of \$100, and you know their willingness to pay for each type of ticket:

Reservation Prices:	Casual Flyer (5)	Frequent Flyer (1)
Basic Economy (MC=100)	\$170	\$210
Standard Economy (MC=100)	\$200	\$300

- a) (2 points) First consider offering only standard economy tickets on this route. What is the profit-maximizing price to charge?

Answer: \$200, profit of 600

- b) (4 points) Now suppose you offer both Basic Economy and Standard Economy. What are the profit-maximizing prices to charge?

Answer: Charge \$170 for BE, earn $70 \times 5 = 350$ from casuals, charge \$260 for standard, earn 160. Total profits are 510.

- c) (4 points) Would it be profitable to offer Basic Economy on this particular route?

Answer: No.

- d) (6 points) This math must be worked out for each individual route. Suppose there were many similar routes of ~1000 miles such as BDL-ORD with the same reservation prices for casual flyers and frequent flyers as this route (the table given above), but that varied in the ratio of casual to frequent flyers. What is the maximum number of casual flyers per frequent flyer in a market such that you would be willing to offer Basic Economy fares to screen consumers?

Answer: pooling profit is $(x+1) \times 100$, while screening profit is $x \times 70 + 160$

$$(x + 1) * 100 = 70x + 160$$

$$100x + 100 = 70x + 160$$

$$30x = 60$$

$$x = 2$$

Problem 6: A vaccine game (25 points)

Five Yale students are going on a foreign trip on which they will live close together. Where they are going, there is a contagious disease which spreads easily among people who live close together. The value of the trip to a student who does not contract the disease is 5. The value of the trip to a student who contracts the disease is 0.

There is a vaccination against the disease. The vaccination costs different amounts for different students (perhaps they have different health plans). Let's call the students 1, 2, 3, 4 and 5 respectively. The vaccination costs 1.1 for student 1, 2.1 for student 2, 3.1 for student 3, 4.1 for student 4, and 4.5 for student 5.

If a student gets vaccinated, she will not contract the disease for sure. But, if she is not vaccinated then her probability of getting the disease depends on the total number in the group who are not vaccinated.

(The probability is higher when more students are not vaccinated.) If she is the only person not to get vaccinated then the probability that she gets the disease is $1/5$ (e.g. from people outside of the group). If there is one other person who is not vaccinated (i.e., two in all including her) then the probability that she gets the disease is $2/5$, etc. Finally if no students get vaccinated at all, she will get the disease for sure.

To make this into a game, suppose that each student aims to maximize her expected payoff and they are risk neutral. If a student chooses to get vaccinated, her payoff is the value of a safe trip minus her cost of vaccination. If a student chooses not to get vaccinated, her expected off is the probability of *not* getting the disease times the value of a safe trip. The students decide, individually and simultaneously, whether or not to get a vaccination.

- a) (4 points) Is it a Nash Equilibrium (NE) that no students choose to get vaccinated? Please explain.
No, it is not a NE.

With no vaccination at all, all students will get the disease. So for each student, they will have an individual incentive to get vaccinated given the benefit 5 is always greater their cost.

- b) (4 points) Is it a NE that all students choose to get vaccinated? Please explain.
No, it is not a NE.

Given all other students get vaccinated, each student has an *individual* incentive to *not* get vaccinated. For example, student 5's payoff is $5 - 4.5 = 0.5$ when she gets vaccinated, and this will increase to $(4/5) * 5 = 4$ if she doesn't get vaccinated.

- c) (7 points) Is it a NE that students 1, 2, 3 get vaccinated but students 4 and 5 do not? Please explain.
No, it is not a NE.

Given the cost structure, we only need to consider students 3 and 4's deviation incentives. Does student 3 want to deviate and not get vaccinated? Her payoff is now $5 - 3.1 = 1.9$, and this will become $(2/5) * 5 = 2$ if she individually switches to not get vaccinated (in which case the probability of being safe drops to $2/5$). Therefore, student 3 will deviate. So the above is not a NE. (If we consider student 4, she actually has no incentive to deviate.)

- d) (10 points) Is it a NE that students 1 and 2 get vaccinated but students 3, 4, 5 do not? Please explain.
Yes, it is a NE.

Given the cost structure, now we only need to consider students 2 and 3's deviation incentives. Does student 2 want to deviate and not get vaccinated? Her payoff is now $5 - 2.1 = 2.9$, and this will become $(1/5) * 5 = 1$ if she does not get vaccinated since the probability of being safe will drop to $1/5$. So student 2 does not want to deviate.

Does student 3 want to deviate and get vaccinated? Her payoff is now $(2/5) * 5 = 2$, and if she deviates and gets vaccinated, her payoff will be $5 - 3.1 = 1.9$. So student 3 does not want to deviate either.

Problem 7: The game of hungry lions (15 points)

A lamb has just been thrown to **four** hungry lions. The lions are chained down in a line, with Lion 1 closest to the lamb and Lion 4 farthest from the lamb. Each lion's chain is short enough that it can only reach the two players immediately adjacent to it. The game proceeds as follows. Lion 1 first decides whether or not to eat the lamb, and then lion 2 decides whether or not to eat Lion 1, and so on. If a lion eats anything, it will be then too heavy to defend itself so become vulnerable to attack. If a lion remains hungry, the next lion will not attack it because a fight of two hungry lions would kill both. Each lion's preferences are fairly natural: best is to eat and stay alive, next best is to stay live but go hungry, next is to eat and be eaten, and worst is to go hungry and be eaten (which wouldn't happen here).

- a) (5 points) Will the lamb get eaten? Please explain.

No, the lamb will survive.

Backward induction: L4 will eat if L3 has eaten; anticipating that, L3 won't eat even if L2 has eaten; anticipating that, L2 will eat if L1 has eaten; then L1 shouldn't eat in the beginning.

- b) (5 points) If there are 5 lions, will your answer to (a) change? Please explain.

Now the lamb will get eaten.

The same backward induction: now L2 won't eat even if L1 has eaten, so anticipating that L1 will eat the lamb.

- c) (5 points) If there are N lions in general (where N is a positive integer), will the lamb get eaten? Please explain.

It depends on whether N is an *odd* number or an *even* number. When it is an even number, we have the same result as in (a) and the lamb will survive. When it is an odd number, we have the same result as in (b) and the lamb will get eaten. [full points once they point out whether N is even or odd matters.]